
Decision on submission to Artificial Intelligence In Medicine

Artificial Intelligence In Medicine <em@editorialmanager.com>
Reply to: Artificial Intelligence In Medicine <support@elsevier.com>
To: Ramanathan Lakshmanan <lramanathan@vit.ac.in>

Sat, 23 May at 2:51 PM

Manuscript Number: **AIIM-D-26-01008**

WavKAN-CL: An Interpretable and Parameter-Efficient Curriculum Learning Framework

for Inter-Patient Arrhythmia Detection

Dear Dr Lakshmanan,

Thank you for submitting your manuscript to Artificial Intelligence In Medicine.

I regret to inform you that the reviewers recommend against publishing your manuscript, and I must therefore reject it. My comments, and any reviewer comments, are below.

For alternative journals that may be more suitable for your manuscript, please refer to our

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We appreciate you submitting your manuscript to Artificial Intelligence In Medicine and thank you for giving us the opportunity to consider your work.

Kind regards,
Carlo Combi, Ph.D.
Editor in Chief

Artificial Intelligence In Medicine

Editor and Reviewer comments:

Reviewer's Responses to Questions

Note: In order to effectively convey your recommendations for improvement to the author(s), and help editors make well-informed and efficient decisions, we ask you to answer the following specific questions about the manuscript and provide additional suggestions where appropriate.

1. Are the objectives and the rationale of the study clearly stated?

Please provide suggestions to the author(s) on how to improve the clarity of the objectives and rationale of the study. Please number each suggestion so that author(s) can more easily respond.

Reviewer #1: YES

2. If applicable, is the application/theory/method/study reported in sufficient detail to allow for its replicability and/or reproducibility?

Please provide suggestions to the author(s) on how to improve the replicability/reproducibility of their study. Please number each suggestion so that the author(s) can more easily respond.

Reviewer #1: N/A

3. If applicable, are statistical analyses, controls, sampling mechanism, and statistical reporting (e.g., P-values, CIs, effect sizes) appropriate and well described?

Please clearly indicate if the manuscript requires additional peer review by a statistician. Kindly provide suggestions to the author(s) on how to improve the statistical analyses, controls, sampling mechanism, or statistical reporting. Please number each suggestion so that the author(s) can more easily respond.

Reviewer #1: YES

4. Could the manuscript benefit from additional tables or figures, or from improving or removing (some of the) existing ones?

Please provide specific suggestions for improvements, removals, or additions of figures or tables. Please number each suggestion so that author(s) can more easily respond.

Reviewer #1: N/A

5. If applicable, are the interpretation of results and study conclusions supported by the data?

Please provide suggestions (if needed) to the author(s) on how to improve, tone down, or expand the study interpretations/conclusions. Please number each suggestion so that the author(s) can more easily respond.

Reviewer #1: N/A

6. Have the authors clearly emphasized the strengths of their study/theory/methods/argument?

Please provide suggestions to the author(s) on how to better emphasize the strengths of their study. Please number each suggestion so that the author(s) can more easily respond.

Reviewer #1: YES

7. Have the authors clearly stated the limitations of their study/theory/methods/argument?

Please list the limitations that the author(s) need to add or emphasize. Please number each limitation so that author(s) can more easily respond.

Reviewer #1: YES

8. Does the manuscript structure, flow or writing need improving (e.g., the addition of subheadings, shortening of text, reorganization of sections, or moving details from one section to another)?

Please provide suggestions to the author(s) on how to improve the manuscript structure and flow. Please number each suggestion so that author(s) can more easily respond.

Reviewer #1: N/A

9. Could the manuscript benefit from language editing?

Reviewer #1: Yes

Reviewer #1: The manuscript titled "WavKAN-CL: An Interpretable and Parameter-Efficient Curriculum Learning Framework for Inter-Patient Arrhythmia Detection" introduces an innovative combination of Wavelet Kolmogorov-Arnold Networks (WavKAN) and a minority-prioritized curriculum learning strategy for arrhythmia detection. The use of Mexican Hat wavelets for morphological feature extraction is a noteworthy improvement over traditional fixed activation functions in CNNs. However, based on the proposed method framework and

contributions of the article, the reviewer believes that the article does not have the potential for publication in AIIM.

Comments:

1. The use of wavelets and the interpretability of the model make it suitable for applications in safety-critical biomedical settings, as it avoids the black-box nature of traditional CNNs.
 2. Overall Macro-F1 Score: The Macro-F1 score of 0.362 is low, especially given the inherent challenges in inter-patient generalization. While the model excels in specific metrics like Ventricular recall, the general performance remains limited. This suggests that the model may struggle with certain types of arrhythmia detection, particularly in minority-class sensitivity.
 3. While the use of a minority-prioritized curriculum learning strategy to tackle class imbalance is effective, it is not a novel approach in the machine learning literature. Curriculum learning itself has been widely explored and is a common technique in addressing class imbalance, especially for rare event detection. The novelty of applying curriculum learning in this context is therefore limited, as similar techniques have been applied in many other fields.
 4. The S-Recall of 0.290 and the F-Recall of 0.028 highlight serious issues with class imbalance, even with the curriculum learning approach. The low performance on these classes indicates that the model may not be clinically reliable for all arrhythmia types.
 5. The zero-shot cross-dataset evaluation on the PTB-XL dataset showed a noticeable drop in performance, further reinforcing the model's limited cross-dataset generalization capability.
 6. The paper lacks detailed real-world testing for deployment on low-power hardware platforms. Despite the claim that the model is lightweight, the lack of post-training quantization and on-device latency benchmarking limits confidence in its practical deployment, particularly for mobile health devices.
- al. If you have any additional suggestions beyond those relevant to the questions above, please number and list them here.

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